# MATHEMATICS — 12th CLASS (PECTAA 2026-27)
# Chapter 1: Graphical Representation of Functions

SOLVED EXERCISES • All 40 Questions Step by Step

Exercise 1.1 • Exercise 1.2 • Exercise 1.3

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## Exercise 1.1 (Solved)

### Q1 (i)

Question. Sketch the graph of the following function using the method of factorization: y = 3(x + 1)(x - 1)

Solution:

Step 1. The factors give the x-intercepts: x + 1 = 0 → x = -1; x - 1 = 0 → x = 1. So the graph cuts the x-axis at (-1, 0) and (1, 0).

Step 2. The vertex is halfway between the intercepts: $x = \frac{-1+1}{2} = 0$. Put x = 0 into the function: y = 3(0 + 1)(0 - 1) = 3(1)(-1) = -3. So the vertex is (0, -3).

Step 3. End behaviour: a = 3 > 0, so the parabola opens upward; $y \to +\infty$ as $x \to \pm\infty$.

Step 4. y-intercept: put x = 0 → y = 3(1)(-1) = -3, so (0, -3) (same as the vertex here).

Step 5. Sketch: upward parabola through (-1, 0), (1, 0), bottom point (0, -3).

**Answer: x-intercepts (-1, 0), (1, 0); vertex (0, -3); y-intercept (0, -3); opens upward, $y \to +\infty$ on both ends.**

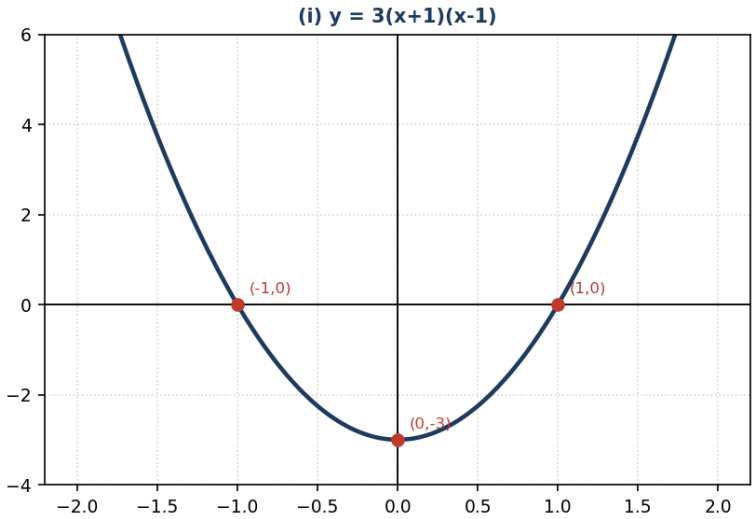



Fig: Exercise 1.1, Q1(i): y = 3(x+1)(x−1); intercepts (−1,0), (1,0); vertex (0,−3).

### Q1 (ii)

Question. Sketch the graph of the following function using the method of factorization: f(x) = -2(x - 1)(x - 2)

Solution:

Step 1. x-intercepts: x - 1 = 0 → x = 1; x - 2 = 0 → x = 2. Points: (1, 0) and (2, 0).

Step 2. Vertex: $x = \frac{1+2}{2} = \frac{3}{2}$. y = -2(³/₂ - 1)(³/₂ - 2) = -2(¹/₂)(-¹/₂) = -2(-¹/₄) = ¹/₂. Vertex: (³/₂, ¹/₂).

Step 3. End behaviour: a = -2 < 0, so the parabola opens downward; $y \to -\infty$ as $x \to \pm\infty$.

Step 4. y-intercept: x = 0 → y = -2(-1)(-2) = -2(2) = -4. Point: (0, -4).

Step 5. Sketch: downward parabola through (1, 0), (2, 0), top point (³/₂, ¹/₂), crossing the y-axis at (0, -4).

**Answer: x-intercepts (1, 0), (2, 0); vertex (³/₂, ¹/₂); y-intercept (0, -4); opens downward, $y \to -\infty$ on both ends.**

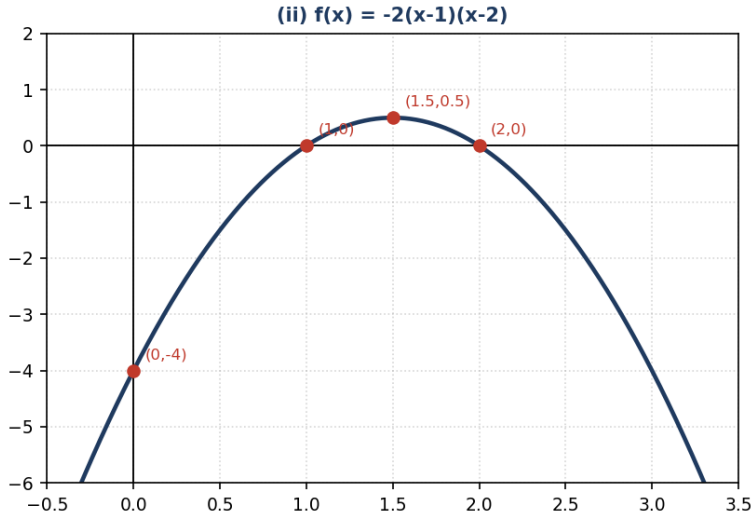



Fig: Exercise 1.1, Q1(ii): f(x) = −2(x−1)(x−2); intercepts (1,0), (2,0); vertex (1.5, 0.5); y-intercept (0,−4).

## Q1 (iii)

Question. Sketch the graph of the following function using the method of factorization: y = (2x - 1)(2x + 1)

Solution:

Step 1. x-intercepts: 2x - 1 = 0 → x = ¹/₂; 2x + 1 = 0 → x = -¹/₂. Points: (-¹/₂, 0) and (¹/₂, 0).

Step 2. Write in the form a(x - h)(x - k): (2x - 1)(2x + 1) = 4(x - ¹/₂)(x + ¹/₂), so a = 4. Vertex: x is the midpoint of -¹/₂ and ¹/₂: -¹/₂ + ¹/₂ = 0, and half of 0 is 0. So x = 0.
$y = (2\cdot 0 - 1)(2\cdot 0 + 1) = (-1)(1) = -1$. Vertex: (0, -1).

Step 3. End behaviour: a = 4 > 0, opens upward; $y \to +\infty$ as $x \to \pm\infty$.

Step 4. y-intercept: x = 0 → y = (-1)(1) = -1, so (0, -1).

Step 5. Sketch: upward parabola through (-¹/₂, 0), (¹/₂, 0), bottom point (0, -1).

**Answer: x-intercepts (-½, 0), (½, 0); vertex (0, -1); y-intercept (0, -1); opens upward, $y \to +\infty$ on both ends.**

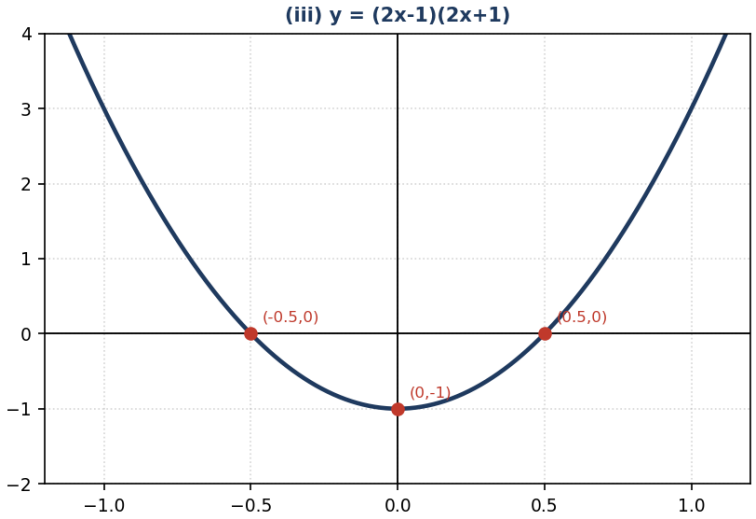



Fig: Exercise 1.1, Q1(iii): y = (2x−1)(2x+1); intercepts (−0.5,0), (0.5,0); vertex (0,−1).

**Q1 (iv)**

Question. Sketch the graph of the following function using the method of factorization: y = 2x² + x - 3

Solution:

Step 1. Factorize first: 2x² + x - 3. We need two numbers with product $2\cdot(-3) = -6$ and sum +1: they are +3 and -2. So 2x² + 3x - 2x - 3 = x(2x + 3) - 1(2x + 3) = (2x + 3)(x - 1).

Step 2. x-intercepts: 2x + 3 = 0 → x = -³/₂; x - 1 = 0 → x = 1. Points: (-³/₂, 0) and (1, 0).

Step 3. Vertex: the x-coordinate is the midpoint of -³/₂ and 1. Add them: -³/₂ + 1 = -³/₂ + ²/₂ = -¹/₂. Half of that: x = -¹/₄. y = 2(-¹/₄)² + (-¹/₄) - 3 = 2(¹/₁₆) - ¹/₄ - 3 = ¹/₈ - ²/₈ - ²⁴/₈ = -²⁵/₈. Vertex: (-¹/₄, -²⁵/₈).

Step 4. End behaviour: a = 2 > 0, opens upward; $y \to +\infty$ as $x \to \pm\infty$.

Step 5. y-intercept: x = 0 → y = -3, so (0, -3).

Step 6. Sketch: upward parabola through (-³/₂, 0), (1, 0), bottom point (-¹/₄, -²⁵/₈), crossing the y-axis at (0, -3).

**Answer: x-intercepts (-³/₂, 0), (1, 0); vertex (-¹/₄, -²⁵/₈); y-intercept (0, -3); opens upward, $y \to +\infty$ on both ends.**

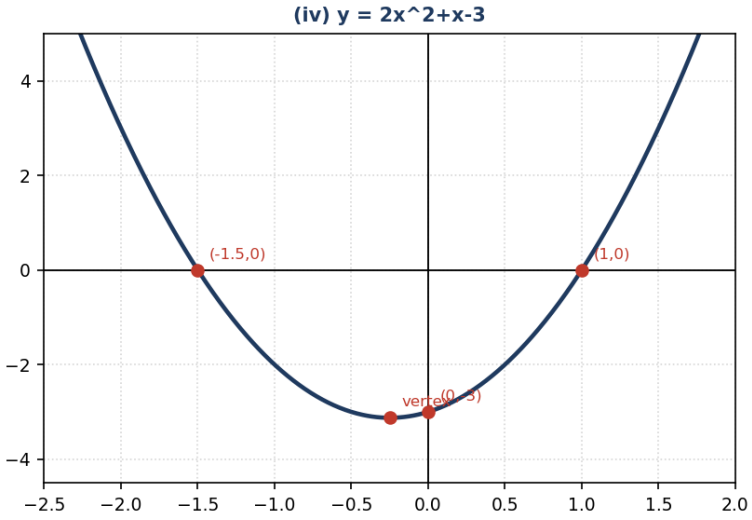



Fig: Exercise 1.1, Q1(iv): y = 2x²+x−3; intercepts (−1.5,0), (1,0); vertex (−0.25,−3.125).

**Q1 (v)**

Question. Sketch the graph of the following function using the method of factorization: f(x) = x² + 2x + 1

Solution:

Step 1. Factorize: x² + 2x + 1 = (x + 1)² — a repeated factor (Case II).

Step 2. The graph touches the x-axis at x = -1. So the vertex is (-1, 0).

Step 3. End behaviour: a = 1 > 0, opens upward; $y \to +\infty$ as $x \to \pm\infty$.

Step 4. y-intercept: x = 0 → y = (0 + 1)² = 1, so (0, 1).

Step 5. Sketch: upward parabola touching the x-axis at (-1, 0), passing through (0, 1) (and symmetrically through (-2, 1)).

**Answer: touches x-axis at (-1, 0) which is the vertex; y-intercept (0, 1); opens upward, $y \to +\infty$ on both ends.**

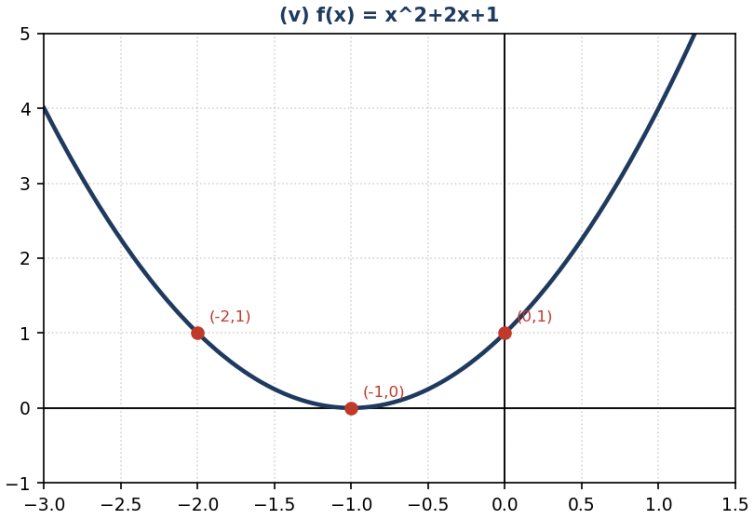


Fig: Exercise 1.1, Q1(v): f(x) = x²+2x+1; touches the x-axis at (−1,0).

### Q1 (vi)

Question. Sketch the graph of the following function using the method of factorization: y = 3(x - 1)²

Solution:

Step 1. Repeated factor (Case II): the graph touches the x-axis at x = 1. Vertex: (1, 0).

Step 2. End behaviour: a = 3 > 0, opens upward; $y \to +\infty$ as $x \to \pm\infty$.

Step 3. y-intercept: x = 0 → y = 3(0 - 1)² = 3(1) = 3, so (0, 3).

Step 4. Sketch: upward parabola touching the x-axis at (1, 0), passing through (0, 3) (and symmetrically through (2, 3)).

**Answer: touches x-axis at (1, 0) which is the vertex; y-intercept (0, 3); opens upward, $y \to +\infty$ on both ends.**

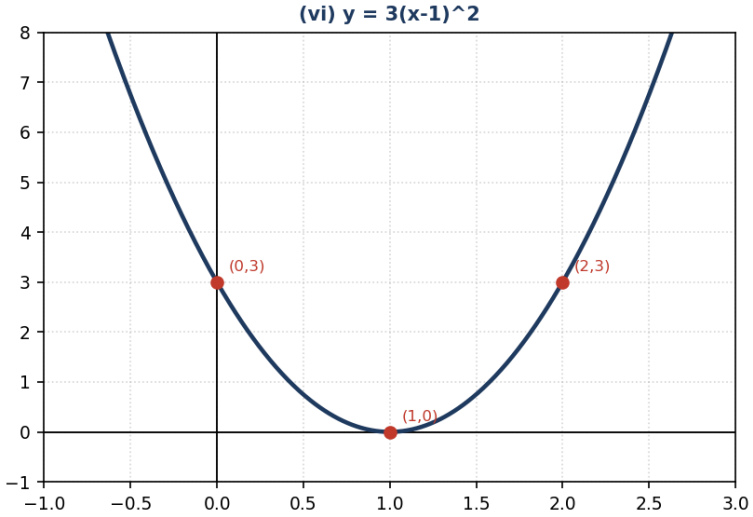



Fig: Exercise 1.1, Q1(vi): y = 3(x−1)²; touches the x-axis at (1,0); y-intercept (0,3).

### Q2 (i)

Question. Find the function whose graph is shown in the following figure: a downward-opening parabola with x-intercepts (-2, 0) and (1, 0), and y-intercept (0, 2).

Solution:

Step 1. The x-intercepts are h = -2 and k = 1, so write the function in factor form: y = a(x - (-2))(x - 1) = a(x + 2)(x - 1).

Step 2. Use the known point (0, 2) to find a: 2 = a(0 + 2)(0 - 1) = a(2)(-1) = -2a.

Step 3. So -2a = 2 → a = -1.

Step 4. Substitute back: y = -1(x + 2)(x - 1) = -(x² + 2x - x - 2) = -(x² + x - 2) = -x² - x + 2.

Step 5. Check: at x = 0, y = 2 ✓; the parabola opens downward ✓.

**Answer: y = -x² - x + 2.**

**Q2 (ii)**

Question. Find the function whose graph is shown in the following figure: an upward-opening parabola that touches the x-axis at (2, 0) (its vertex) and passes through (1, 1).

Solution:

Step 1. The graph touches the x-axis at x = 2, so use the repeated-factor form: $y = a(x - 2)^2$.

Step 2. Use the known point (1, 1): $1 = a(1 - 2)^2 = a(-1)^2 = a$.

Step 3. So a = 1.

Step 4. Substitute back: $y = (x - 2)^2 = x^2 - 4x + 4$.

Step 5. Check: at x = 2, y = 0 ✓; at x = 1, y = 1 ✓; opens upward ✓.

**Answer: $y = (x - 2)^2$ (i.e. $y = x^2 - 4x + 4$).**

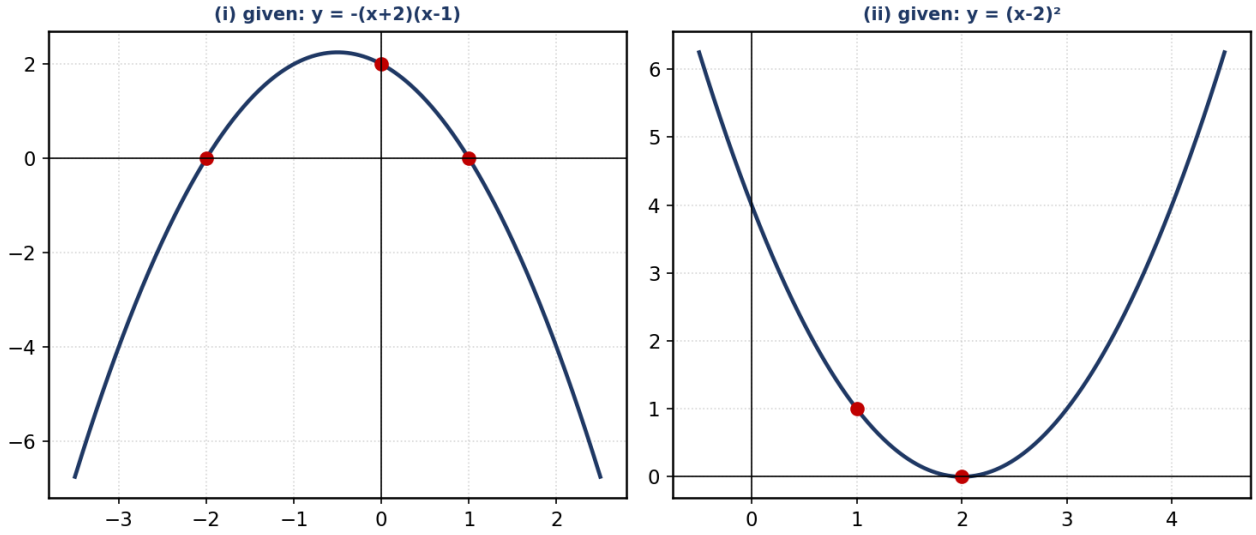



Fig: Exercise 1.1, Q2: the given graphs whose equations are found in the solution.

**Q3**

Question. Find an equation of the graph of a parabola whose x-intercepts are x = -1 and x = -2 and which passes through the point (-3, 2).

Solution:

Step 1. x-intercepts -1 and -2 give the factor form: $y = a(x + 1)(x + 2)$.

Step 2. Substitute the point (-3, 2): $2 = a(-3 + 1)(-3 + 2) = a(-2)(-1) = 2a$.

Step 3. So $2a = 2 \rightarrow a = 1$.

Step 4. Substitute back: $y = (x + 1)(x + 2) = x^2 + 3x + 2$.

Step 5. Check: at x = -3, y = 9 - 9 + 2 = 2 ✓; intercepts at -1, -2 ✓.

**Answer: $y = x^2 + 3x + 2$.**

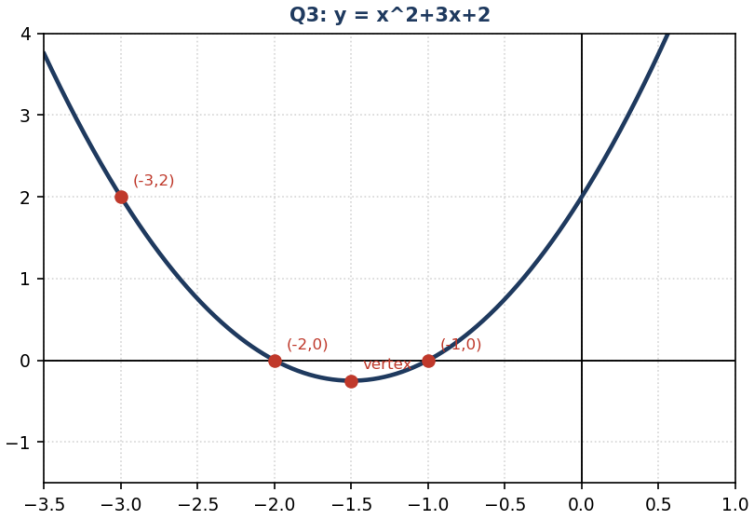



Fig: Exercise 1.1, Q3: y = x²+3x+2; intercepts (−1,0), (−2,0); passes through (−3,2).

**Q4**

Question. Find an equation of the graph of a parabola which touches the x-axis at x = -1 and cuts the y-axis at y = -2.

Solution:

Step 1. "Touches the x-axis at x = -1" means a repeated factor: y = a(x + 1)².

Step 2. "Cuts the y-axis at y = -2" means the y-intercept is (0, -2). Substitute x = 0, y = -2: -2 = a(0 + 1)² = a(1) = a.

Step 3. So a = -2.

Step 4. Substitute back: y = -2(x + 1)² = -2(x² + 2x + 1) = -2x² - 4x - 2.

Step 5. Check: at x = 0, y = -2 ✓; touches x-axis at x = -1 ✓; opens downward (a < 0) ✓.

**Answer: y = -2(x + 1)² (i.e. y = -2x² - 4x - 2).**

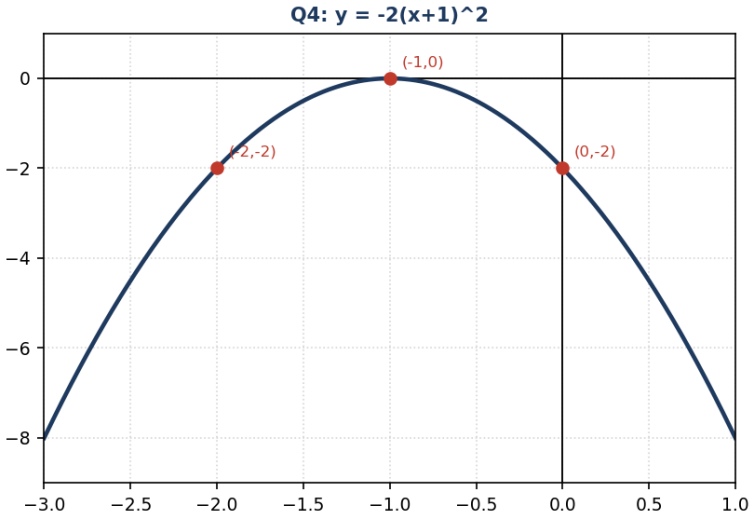



Fig: Exercise 1.1, Q4: y = −2(x+1)²; touches the x-axis at (−1,0); y-intercept (0,−2).

### Q5

Question. Find an equation of a quadratic function whose graph cuts the x-axis at x = -1 and x = 1, and whose vertex is at (0, -2).

Solution:

Step 1. x-intercepts -1 and 1 give the factor form: y = a(x + 1)(x - 1) = a(x² - 1).

Step 2. The vertex is (0, -2); substitute x = 0, y = -2: -2 = a(0 + 1)(0 - 1) = a(1)(-1) = -a.

Step 3. So -a = -2 → a = 2.

Step 4. Substitute back: y = 2(x + 1)(x - 1) = 2x² - 2.

Step 5. Check: x-intercepts: 2x² - 2 = 0 → $x = \pm 1$ ✓; vertex: midpoint x = 0, y = -2 ✓; opens upward ✓.

**Answer: y = 2x² - 2.**

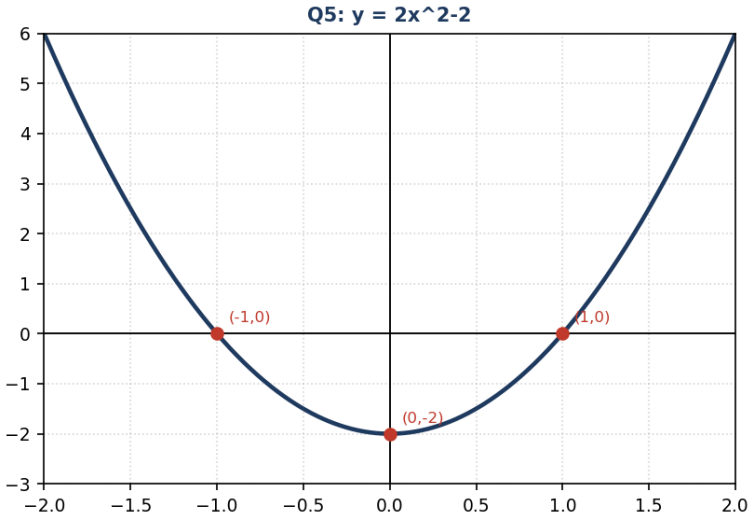



Fig: Exercise 1.1, Q5: y = 2x²−2; intercepts (−1,0), (1,0); vertex (0,−2).

## Exercise 1.2 (Solved)

### Q1

Question. Which of the following functions are algebraic?

$f(x) = \sec x$, $g(x) = \frac{2x-3}{x+1}$, $h(x) = \sqrt{x^2+2x}$, $k(x) = \frac{e^x - e^{-x}}{2}$, r(x) = (x - 1)³

Solution: A function is algebraic if it can be built from x using only addition, subtraction, multiplication, division and roots.

Step 1. $f(x) = \sec x$ — contains a trigonometric function; cannot be built by algebraic operations → not algebraic (transcendental).

Step 2. $g(x) = \frac{2x-3}{x+1}$ — only multiplication, subtraction, division of polynomials → algebraic.

Step 3. $h(x) = \sqrt{x^2 + 2x}$ — square root of a polynomial → algebraic.

Step 4. $k(x) = \frac{e^x - e^{-x}}{2} = \sinh x$ — contains e^x → not algebraic (transcendental).

Step 5. r(x) = (x - 1)³ — only subtraction and multiplication → algebraic.

**Answer: The algebraic functions are $g(x) = \frac{2x-3}{x+1}$, $h(x) = \sqrt{x^2 + 2x}$, and r(x) = (x - 1)³.**

**Q2**

Question. Which of the following functions are transcendental?

$f(x) = \frac{1}{x}$, $g(x) = \cot x$, $h(x) = e^{2x} + \sqrt{x^2 + 2x}$, $k(x) = \frac{e^x + e^{-x}}{2}$, $r(x) = (x-1)^{-3/2}$

Solution: A function is transcendental if it is NOT algebraic (i.e. it cannot be built from x using only +, -, ×, ÷, roots).

Step 1. $f(x) = \frac{1}{x}$ — only division → algebraic, so not transcendental.

Step 2. $g(x) = \cot x$ — trigonometric → transcendental.

Step 3. $h(x) = e^{2x} + \sqrt{x^2 + 2x}$ — contains $e^{2x}$ (the algebraic part $\sqrt{x^2+2x}$ cannot cancel it) → transcendental.

Step 4. $k(x) = \frac{e^x + e^{-x}}{2} = \cosh x$ — contains e^x → transcendental.

Step 5. $r(x) = (x-1)^{-3/2}$ — a root/power of a polynomial ($\frac{1}{(x-1)^{3/2}}$) → algebraic, so not transcendental.

**Answer: The transcendental functions are $g(x) = \cot x$, $h(x) = e^{2x} + \sqrt{x^2 + 2x}$, and $k(x) = \frac{e^x + e^{-x}}{2}$.**

**Q3**

Question. Discuss domain, range, and graph of the function f(x) = 3^x.

Solution:

Step 1. Domain: 3^x is defined for every real x → Domain = $(-\infty, \infty)$.

Step 2. Positivity and range: 3^x > 0 for all real x; as $x \to +\infty$, $3^x \to +\infty$ and as $x \to -\infty$, $3^x \to 0^+$. Since the function is continuous, it takes every positive value → Range = $(0, \infty)$.

Step 3. Intercepts: y-intercept: x = 0 → y = 3⁰ = 1, so (0, 1). No x-intercept (3^x > 0 always).

Step 4. Behaviour: base 3 > 1, so f is strictly increasing (exponential growth); the line y = 0 is the horizontal asymptote (as $x \to -\infty$).

Step 5. Graph: starts near y = 0 on the far left, passes through (0, 1), (-1, ⅓), (1, 3), (2, 9), and rises steeply to the right.

**Answer: Domain $(-\infty, \infty)$; Range $(0, \infty)$; y-intercept (0, 1); strictly increasing; horizontal asymptote y = 0.**

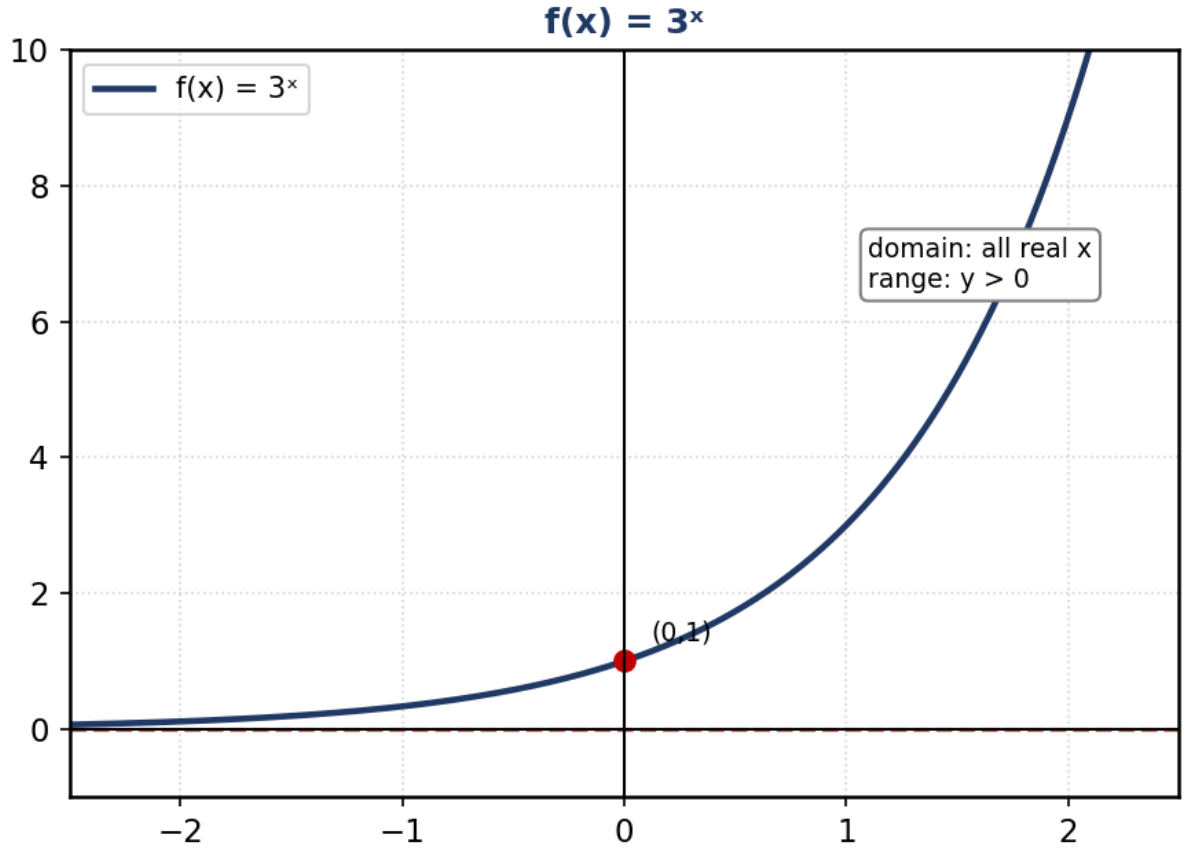



Fig: f(x) = 3ˣ: domain all real x, range y > 0.

**Q4**

Question. Discuss domain, range, and graph of the function f(x) = (¹/₃)^x.

Solution:

Step 1. Domain: (¹/₃)^x is defined for every real x → Domain = $(-\infty, \infty)$.

Step 2. Positivity and range: (¹/₃)^x > 0 for all real x; as $x \to +\infty$, $(\frac{1}{3})^x \to 0^+$ and as $x \to -\infty$, $(\frac{1}{3})^x \to +\infty$. The function is continuous → Range = $(0, \infty)$.

Step 3. Intercepts: y-intercept: x = 0 → y = (¹/₃)⁰ = 1, so (0, 1). No x-intercept.

Step 4. Behaviour: base ¹/₃ is between 0 and 1, so f is strictly decreasing (exponential decay); the line y = 0 is the horizontal asymptote (as $x \to +\infty$).

Step 5. Graph: starts very high on the far left, falls through (-1, 3), (0, 1), (1, ¹/₃), (2, ¹/₉), and flattens toward y = 0 on the right.

**Answer: Domain $(-\infty, \infty)$; Range $(0, \infty)$; y-intercept (0, 1); strictly decreasing; horizontal asymptote y = 0.**

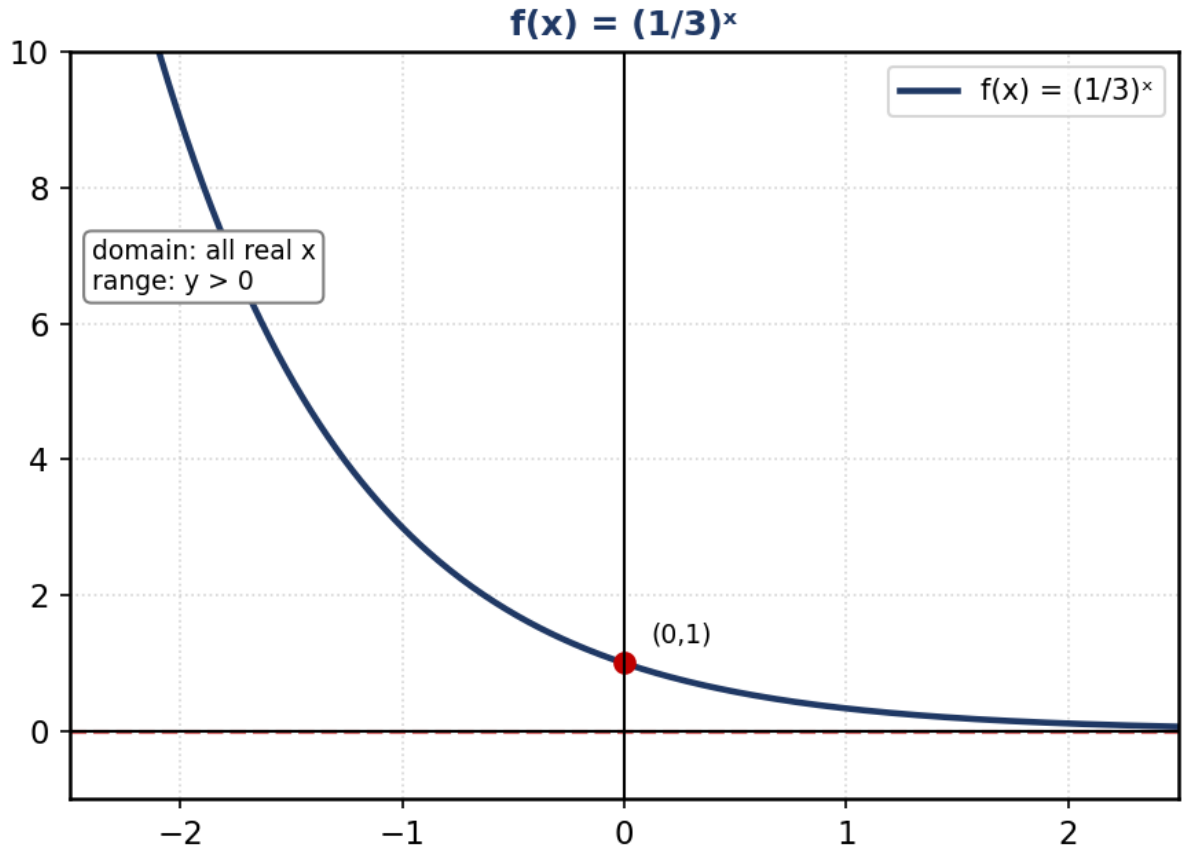



Fig: f(x) = (1/3)ˣ: domain all real x, range y > 0.

**Q5 (i)**

Question. Prove the following hyperbolic identity: $1 - \tanh^2 x = \operatorname{sech}^2 x$

Solution:

Step 1. Start from the left side and write $\tanh x$ and $\operatorname{sech} x$ with $\sinh x$ and $\cosh x$: $\tanh x = \dfrac{\sinh x}{\cosh x}$, $\operatorname{sech} x = \dfrac{1}{\cosh}$.

Step 2. LHS $= 1 - \tanh^2 x = 1 - \dfrac{\sinh^2 x}{}$.

Step 3. Write 1 as $\dfrac{\cosh^2 x}{\cosh^2 x}$: LHS $= \dfrac{\cosh^2 x - \sinh^2 x}{}$.

Step 4. Use the key identity $\cosh^2 x - \sinh^2 x = 1$: LHS $= \dfrac{1}{\cosh^2 x}$.

Step 5. $\dfrac{1}{\cosh x} = \operatorname{sech} x$, so $\dfrac{1}{\cosh^2 x} = \operatorname{sech}^2 x$ = RHS.

**Answer:** $1 - \tanh^2 x = \operatorname{sech}^2 x$. ■

**Q5 (ii)**

Question. Prove the following hyperbolic identity: $\operatorname{csch}^2 x = \coth^2 x - 1$

Solution:

Step 1. Start from the right side: $\coth x = \dfrac{\cosh x}{}$, so RHS $= \dfrac{\cosh^2 x}{\sinh^2 x} - 1$.

Step 2. Write 1 as $\dfrac{\sinh^2 x}{\sinh^2 x}$: RHS $= \dfrac{\cosh^2 x - \sinh^2 x}{\sinh^2 x}$.

Step 3. Use $\cosh^2 x - \sinh^2 x = 1$: RHS $= \frac{1}{\text{h}^2 x}$.

Step 4. $\frac{1}{\sinh x} = \operatorname{csch} x$, so $\frac{1}{\sinh^2 x} = \operatorname{csch}^2 x =$ LHS.

**Answer:** $\operatorname{csch}^2 x = \coth^2 x - 1$. ■

**Q5 (iii)**

Question. Prove the following hyperbolic identity: $\sinh 2x = 2 \sinh x \cosh x$

Solution:

Step 1. Start from the right side and use the exponential definitions: $\sinh x = \frac{e^x - e^{-x}}{2}$, $\cosh x = \frac{e^x + e^{-x}}{2}$.

Step 2. RHS $= 2 \cdot \frac{e^x - e^{-x}}{2} \cdot \frac{e^x + e^{-x}}{2} = \frac{2(e^x - e^{-x})(e^x + e^{-x})}{4}$.

Step 3. Simplify: $^2/_4 = {}^1/_2$, and $(e^x - e^{-x})(e^x + e^{-x}) = (e^x)^2 - (e^{-x})^2 = e^{2x} - e^{-2x}$ (difference of squares).

Step 4. So RHS $= \frac{e^{2x} - e^{-2x}}{2}$.

Step 5. By definition, $\frac{e^{2x} - e^{-2x}}{2} = \sinh 2x =$ LHS.

**Answer:** $\sinh 2x = 2 \sinh x \cosh x$. ■

**Q5 (iv)**

Question. Prove the following hyperbolic identity: $\cosh 2x = \cosh^2 x + \sinh^2 x$

Solution:

Step 1. Start from the right side and use the exponential definitions.

Step 2. $\cosh^2 x = (\frac{e^x + e^{-x}}{2})^2 = \frac{e^{2x} + 2 \cdot e^x \cdot e^{-x} + e^{-2x}}{4} = \frac{e^{2x} + 2 + e^{-2x}}{4}$.

Step 3. $\sinh^2 x = (\frac{e^x - e^{-x}}{2})^2 = \frac{e^{2x} - 2 \cdot e^x \cdot e^{-x} + e^{-2x}}{4} = \frac{e^{2x} - 2 + e^{-2x}}{4}$.

Step 4. Add them: RHS $= \frac{(e^{2x} + 2 + e^{-2x}) + (e^{2x} - 2 + e^{-2x})}{4} = \frac{2e^{2x} + 2e^{-2x}}{4}$.

Step 5. Simplify: $\frac{2e^{2x} + 2e^{-2x}}{4} = \frac{e^{2x} + e^{-2x}}{2} = \cosh 2x =$ LHS.

**Answer:** $\cosh 2x = \cosh^2 x + \sinh^2 x$. ■

**Q6 (i)**

Question. Prove the following inverse hyperbolic formula: $\tanh^{-1} x = \frac{1}{2} \ln(\frac{1+x}{1-x})$

Solution:

Step 1. Let $y = \tanh^{-1} x$, so that $x = \tanh y = \frac{e^y - e^{-y}}{e^y + e^{-y}}$.

Step 2. Cross-multiply: $x(e^y + e^{-y}) = e^y - e^{-y} \rightarrow x \cdot e^y + x \cdot e^{-y} = e^y - e^{-y}$.

Step 3. Collect e^y terms on one side and $e^{-y}$ terms on the other: $x \cdot e^{-y} + e^{-y} = e^y - x \cdot e^y \rightarrow e^{-y}(x+1) = e^y(1-x)$.

Step 4. Divide both sides by $e^{-y}(1-x)$: $e^{2y} = \frac{1+x}{1-x}$.

Step 5. Take ln of both sides: $2y = \ln(\frac{1+x}{1-x})$.

Step 6. So $y = \frac{1}{2} \cdot \ln(\frac{1+x}{1-x})$, i.e. $\tanh^{-1} x = \frac{1}{2} \ln(\frac{1+x}{1-x})$.

**Answer:** $\tanh^{-1} x = \frac{1}{2}\ln(\frac{1+x}{1-x})$. ■

**Q6 (ii)**

Question. Prove the following inverse hyperbolic formula: $\coth^{-1} x = \frac{1}{2}\ln(\frac{x+1}{x-1})$

Solution:

Step 1. Let $y = \coth^{-1} x$, so that $x = \coth y = \frac{e^y + e^{-y}}{e^y - e^{-y}}$.

Step 2. Cross-multiply: $x(e^y - e^{-y}) = e^y + e^{-y} \rightarrow x\cdot e^y - x\cdot e^{-y} = e^y + e^{-y}$.

Step 3. Collect: $x\cdot e^y - e^y = x\cdot e^{-y} + e^{-y} \rightarrow e^y(x-1) = e^{-y}(x+1)$.

Step 4. Divide both sides by $e^{-y}(x-1)$: $e^{2y} = \frac{x+1}{x-1}$.

Step 5. Take ln of both sides: $2y = \ln(\frac{x+1}{x-1})$.

Step 6. So $y = \frac{1}{2}\cdot\ln(\frac{x+1}{x-1})$, i.e. $\coth^{-1} x = \frac{1}{2}\ln(\frac{x+1}{x-1})$.

**Answer:** $\coth^{-1} x = \frac{1}{2}\ln(\frac{x+1}{x-1})$. ■

**Q6 (iii)**

Question. Prove the following inverse hyperbolic formula: $\operatorname{sech}^{-1} x = \ln(\frac{1+\sqrt{1-x^2}}{x})$

Solution:

Step 1. Let $y = \operatorname{sech}^{-1} x$, so that $x = \operatorname{sech} y = \frac{2}{e^y + e^{-y}}$.

Step 2. Cross-multiply: $x(e^y + e^{-y}) = 2 \rightarrow x\cdot e^y + x\cdot e^{-y} = 2$.

Step 3. Multiply through by e^y: x(e^y)² + x = 2e^y → x(e^y)² - 2e^y + x = 0 — a quadratic in e^y.

Step 4. Apply the quadratic formula: $e^y = \frac{2 \pm \sqrt{4-4x^2}}{2x} = \frac{2 \pm 2\sqrt{1-x^2}}{2x} = \frac{1 \pm \sqrt{1-x^2}}{x}$.

Step 5. Choose the sign: $\operatorname{sech}^{-1} x$ is defined for $x \in (0, 1]$ with range $y \geq 0$, so we need $e^y \geq 1$. Check the minus sign: $\frac{1-\sqrt{1-x^2}}{x} \leq 1$ for $x \in (0, 1]$ (it equals 1 only at x = 1 and drops toward 0 as $x \rightarrow 0^+$), which would give $y \leq 0$ — wrong branch. The plus sign gives $\frac{1+\sqrt{1-x^2}}{x} \geq 1$ ✓.

Step 6. Take the plus sign and apply ln: $y = \ln(\frac{1+\sqrt{1-x^2}}{x})$.

**Answer:** $\operatorname{sech}^{-1} x = \ln(\frac{1+\sqrt{1-x^2}}{x})$. ■

**Q6 (iv)**

Question. Prove the following inverse hyperbolic formula: $\operatorname{csch}^{-1} x = \ln(\frac{1}{x} + \sqrt{\frac{1}{x^2}+1})$

Solution:

Step 1. Let $y = \operatorname{csch}^{-1} x$, so that $x = \operatorname{csch} y = \frac{2}{e^y - e^{-y}}$.

Step 2. Cross-multiply: $x(e^y - e^{-y}) = 2 \rightarrow x\cdot e^y - x\cdot e^{-y} = 2$.

Step 3. Multiply through by e^y: x(e^y)² - x = 2e^y → x(e^y)² - 2e^y - x = 0 — a quadratic in e^y.

Step 4. Apply the quadratic formula: $e^y = \frac{2 \pm \sqrt{4+4x^2}}{2x} = \frac{2 \pm 2\sqrt{1+x^2}}{2x} = \frac{1 \pm \sqrt{1+x^2}}{x}$.

Step 5. Choose the sign so that e^y > 0 (an exponential is always positive). Since $\sqrt{1+x^2} > 1$, the numerator $1 - \sqrt{1+x^2}$ is negative, so the minus-sign candidate below is positive only when x

< 0:

$$\frac{1-\sqrt{1+x^2}}{x}$$

and the plus-sign candidate below is positive only when x > 0:

$$\frac{1+\sqrt{1+x^2}}{x}$$

The single formula in the question handles both cases at once, because:

$$\sqrt{\frac{1}{x^2}+1}=\frac{\sqrt{1+x^2}}{|x|}$$

Step 6. Hence $y=\ln(\frac{1}{x}+\sqrt{\frac{1}{x^2}+1})$.

**Answer:** $\operatorname{csch}^{-1}x=\ln(\frac{1}{x}+\sqrt{\frac{1}{x^2}+1})$. ■

## Exercise 1.3 (Solved)

### Q1 (i)

Question. Let $f(x)=\log_4(x-2)$. (i) Show that f(x) is a one-to-one function using the horizontal line test.

Solution:

Step 1. The graph of $f(x)=\log_4(x-2)$ is the graph of $\log_4 x$ shifted 2 units to the right. Since the base 4 > 1, $\log_4 x$ is strictly increasing, and shifting does not change this.

Step 2. A strictly increasing function takes every value exactly once, so every horizontal line meets the graph at most once — it passes the horizontal line test.

Step 3. Hence f is one-to-one. (Algebraic check: if $\log_4(a-2)=\log_4(b-2)$, raising 4 to both sides gives a - 2 = b - 2, so a = b.)

**Answer: f is strictly increasing, so it passes the horizontal line test and is one-to-one.**

### Q1 (ii)

Question. Let $f(x)=\log_4(x-2)$. (ii) Find the inverse function f⁻¹(x) algebraically.

Solution:

Step 1. Set $y=\log_4(x-2)$ and swap x and y: $x=\log_4(y-2)$.

Step 2. Convert to exponential form: 4^x = y - 2.

Step 3. Solve for y: y = 4^x + 2.

Step 4. So f⁻¹(x) = 4^x + 2.

**Answer: f⁻¹(x) = 4^x + 2.**

### Q1 (iii)

Question. Let $f(x)=\log_4(x-2)$. (iii) State the domain and range of f(x) and f⁻¹(x).

Solution:

Step 1. $f(x) = \log_4(x-2)$: needs x - 2 > 0 → Domain = $(2, \infty)$; a logarithm can output any real number → Range = $(-\infty, \infty)$.

Step 2. $f^{-1}(x) = 4^x + 2$: an exponential eats all real x → Domain = $(-\infty, \infty)$; since $4^x > 0$, $4^x + 2 > 2$ → Range = $(2, \infty)$.

Step 3. Check the swap rule: Dom f = Range $f^{-1}$ = $(2, \infty)$ ✓ and Range f = Dom $f^{-1}$ = $(-\infty, \infty)$ ✓.

**Answer: Dom f = $(2, \infty)$, Range f = $(-\infty, \infty)$; Dom $f^{-1}$ = $(-\infty, \infty)$, Range $f^{-1}$ = $(2, \infty)$.**

## Q1 (iv)

Question. Let $f(x) = \log_4(x-2)$. (iv) Verify graphically that the graphs of f(x) and $f^{-1}(x)$ are symmetric about the line y = x.

Solution:

Step 1. Plot both curves together with the line y = x.

Step 2. Pick points on f: (3, 0) since $\log_4(3-2) = \log_4 1 = 0$; (6, 1) since $\log_4 4 = 1$; (18, 2) since $\log_4 16 = 2$.

Step 3. The matching points on $f^{-1}(x) = 4^x + 2$ are (0, 3), (1, 6), (2, 18) — each is the mirror image (coordinates swapped).

Step 4. Every point (a, b) on f has its mirror (b, a) on $f^{-1}$, so the two graphs are symmetric (mirror images) about the line y = x ✓.

**Answer: Verified — each point (a, b) on f reflects to (b, a) on $f^{-1}$ across y = x.**

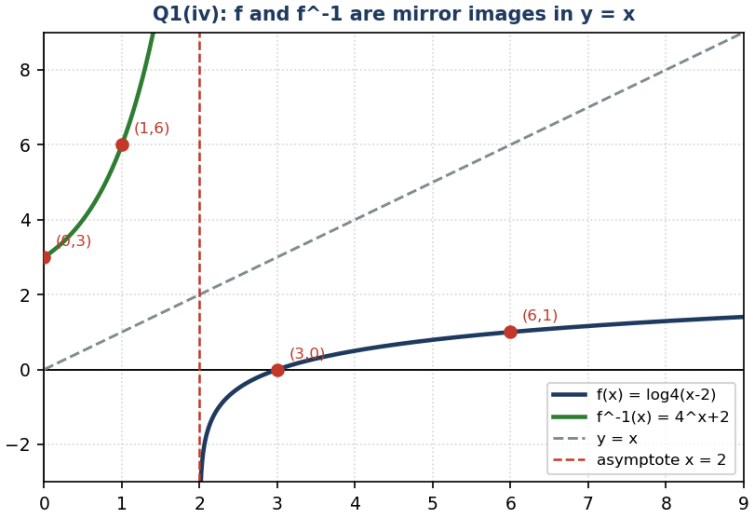



Fig: Exercise 1.3, Q1(iv): f(x) = $\log_4(x-2)$ and $f^{-1}(x) = 4^x+2$ are mirror images in the line y = x.

## Q2 (i)

Question. Let $f(x) = \sin x$ for all $x \in [\frac{-\pi}{2}, \frac{\pi}{2}]$. (i) State the domain and range of f(x) and $f^{-1}(x)$.

Solution:

Step 1. $f(x) = \sin x$ on $[\frac{-\pi}{2}, \frac{\pi}{2}]$: Domain = $[\frac{-\pi}{2}, \frac{\pi}{2}]$.

Step 2. $\sin(\frac{-\pi}{2}) = -1$, $\sin(\frac{\pi}{2}) = 1$, and $\sin$ rises steadily between → Range = [-1, 1].

Step 3. Swapping domain and range for the inverse: Dom $f^{-1}$ = Range f = [-1, 1]; Range $f^{-1}$ = Dom f = $[\frac{-\pi}{2}, \frac{\pi}{2}]$.

**Answer: Dom f = $[\frac{-\pi}{2}, \frac{\pi}{2}]$, Range f = [-1, 1]; Dom $f^{-1}$ = [-1, 1], Range $f^{-1}$ = $[\frac{-\pi}{2}, \frac{\pi}{2}]$.**

## Q2 (ii)

Question. Let $f(x) = \sin x$ for all $x \in [\frac{-\pi}{2}, \frac{\pi}{2}]$. (ii) Sketch the graph of f(x) and show that it is one-to-one using the horizontal line test.

Solution:

Step 1. The graph starts at $(\frac{-\pi}{2}, -1)$, passes through (0, 0), and ends at $(\frac{\pi}{2}, 1)$, rising all the way ($\sin$ is strictly increasing on this interval).

Step 2. Because it never turns back, any horizontal line cuts it at most once.

Step 3. So it passes the horizontal line test → f is one-to-one.

**Answer: The graph rises steadily from $(\frac{-\pi}{2}, -1)$ to $(\frac{\pi}{2}, 1)$; passes the horizontal line test, hence one-to-one.**

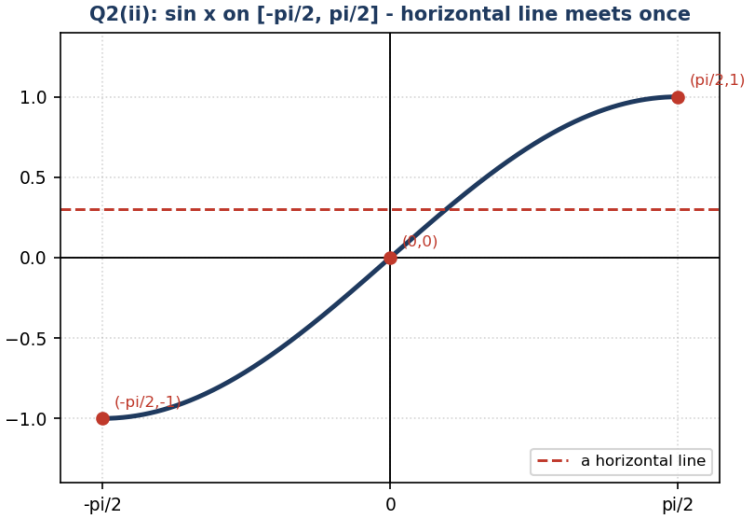



Fig: Exercise 1.3, Q2(ii): sin x on [−π/2, π/2]; any horizontal line meets the graph exactly once.

## Q2 (iii)

Question. Let $f(x) = \sin x$ for all $x \in [\frac{-\pi}{2}, \frac{\pi}{2}]$. (iii) Draw the graph of $f^{-1}(x)$ using the reflection of f(x) along the line y = x.

Solution:

Step 1. Reflect each point across y = x (swap coordinates): $(\frac{-\pi}{2}, -1) \to (-1, \frac{-\pi}{2})$; $(0, 0) \to (0, 0)$; $(\frac{\pi}{2}, 1) \to (1, \frac{\pi}{2})$.

Step 2. Joining these gives a rising curve from $(-1, \frac{-\pi}{2})$ to $(1, \frac{\pi}{2})$ — the graph of $f^{-1}(x) = \sin^{-1} x$.

**Answer: f⁻¹ is the mirror image of f across y = x (the arcsine curve through $(-1, \frac{-\pi}{2})$, (0, 0), $(1, \frac{\pi}{2})$).**

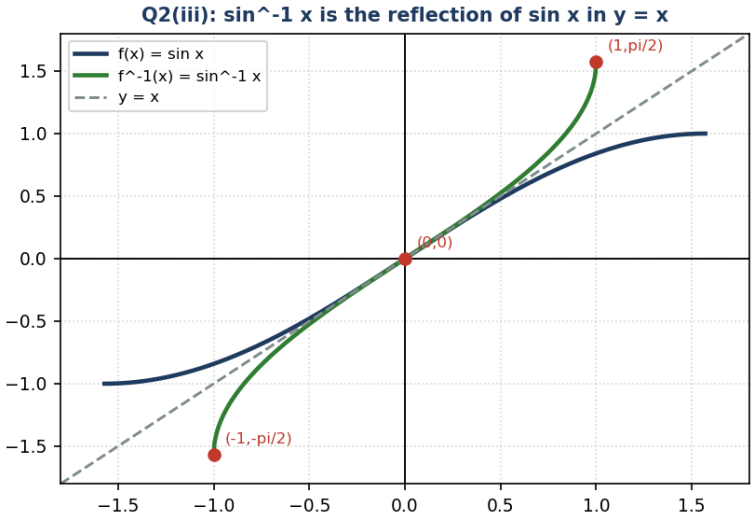



Fig: Exercise 1.3, Q2(iii): sin⁻¹x is the reflection of sin x in the line y = x.

**Q3**

Question. A one-to-one function is defined by f(x) = (x - 1)³, $x \in [0, 3]$. Using the concept of reflection along the line y = x, sketch the graph of the inverse function f⁻¹(x).

Solution:

Step 1. First find f⁻¹ algebraically: y = (x - 1)³ → swap: x = (y - 1)³ → take cube root: $y - 1 = \sqrt[3]{x}$ → $f^{-1}(x) = 1 + \sqrt[3]{x}$.

Step 2. Domain/range: f(0) = -1, f(3) = 8, and f is strictly increasing, so Dom f = [0, 3], Range f = [-1, 8]. Hence Dom f⁻¹ = [-1, 8], Range f⁻¹ = [0, 3].

Step 3. Sketch f: curve from (0, -1) through (1, 0) to (3, 8).

Step 4. Reflect across y = x (swap coordinates): $(0, -1) \to (-1, 0)$; $(1, 0) \to (0, 1)$; $(3, 8) \to (8, 3)$. Joining these reflected points gives the graph of $f^{-1}(x) = 1 + \sqrt[3]{x}$, a curve rising from (-1, 0) through (0, 1) to (8, 3).

**Answer: $f^{-1}(x) = 1 + \sqrt[3]{x}$; its graph is the reflection of f across the line y = x, passing through (-1, 0), (0, 1), (8, 3).**

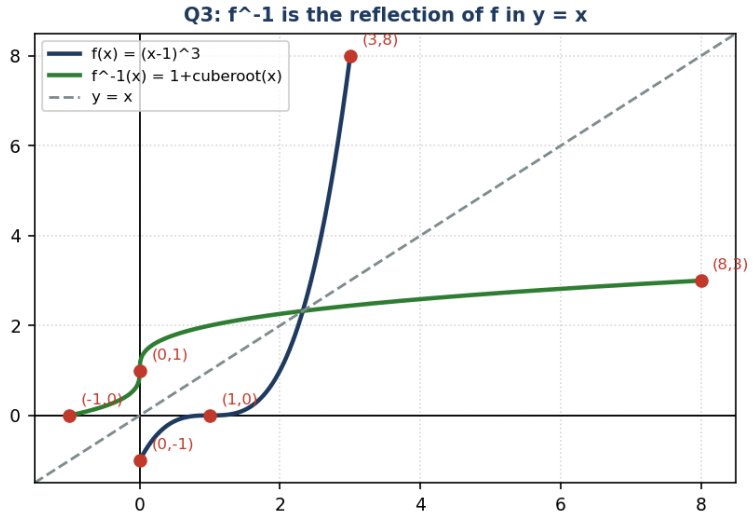



Fig: Exercise 1.3, Q3: $f^{-1}(x) = 1+\sqrt[3]{x}$ is the reflection of $f(x) = (x-1)^3$ in the line y = x.

**Q4 (i)**

Question. Given the function $f(x) = \sqrt{x}$. (i) Find the equation of the function after a vertical stretch by factor 2.

Solution:

Step 1. Vertical stretch rule: $y = a \cdot f(x)$ with a = 2.

Step 2. So y = $2\sqrt{x}$.

**Answer:** $y = 2\sqrt{x}$.

**Q4 (ii)**

Question. Given the function $f(x) = \sqrt{x}$. (ii) Find the equation of the function after a horizontal shift 3 units to the right.

Solution:

Step 1. Horizontal shift rule: y = f(x - h) with h = 3.

Step 2. So y = $\sqrt{x-3}$.

**Answer:** $y = \sqrt{x-3}$.

**Q4 (iii)**

Question. Given the function $f(x) = \sqrt{x}$. (iii) Find the equation after applying both transformations (i) and (ii).

Solution:

Step 1. Start from part (i): $y = 2\sqrt{x}$.

Step 2. Now shift 3 units right: replace x by x - 3 → y = $2\sqrt{x-3}$.

**Answer:** $y = 2\sqrt{x-3}$.

**Q5**

Question. Given the function $f(x) = \frac{1}{x}$, $x \neq 0$. Find the equation of the transformed function after applying the following transformations in order: horizontal shift 2 units to the left, horizontal compression by factor 2, vertical stretching by factor 3, and vertical shift 1 unit upward.

Solution: Apply each transformation to the current function, in order.

Step 1. Shift 2 units left: replace x by x + 2: $y = \frac{1}{x+2}$. Call this g(x).

Step 2. Horizontal compression by factor 2: replace x by 2x in g: $y = g(2x) = \frac{1}{2x+2}$.

Step 3. Vertical stretch by factor 3: multiply by 3: $y = \frac{3}{2x+2}$.

Step 4. Shift 1 unit upward: add 1: $y = \frac{3}{2x+2} + 1$.

**Answer:** $y = \frac{3}{2x+2} + 1$.

**Q6**

Question. Given the function $f(x) = \ln x$, find the equation of the transformed function obtained after shifting the graph 1 unit to the right, vertically compressing it by a factor of 2, and then shifting it 3 units upward.

Solution:

Step 1. Shift 1 unit right: $y = \ln(x-1)$.

Step 2. Vertical compression by factor 2: multiply by ½: $y = \frac{1}{2}\cdot\ln(x-1)$.

Step 3. Shift 3 units upward: add 3: $y = \frac{1}{2}\cdot\ln(x-1) + 3$.

**Answer:** $y = \frac{1}{2}\cdot\ln(x-1) + 3$.

**Q7**

Question. Given the function $f(x) = \sqrt{\frac{2x-1}{x+3}}$. Find the equation of the transformed function after applying in order: the horizontal shift 2 units to the left, horizontal compression by factor 3, vertical compression by factor 2, and vertical shift 1 unit downward.

Solution:

Step 1. Shift 2 units left: replace x by x + 2: $y = \sqrt{\frac{2(x+2)-1}{(x+2)+3}} = \sqrt{\frac{2x+4-1}{x+5}} = \sqrt{\frac{2x+3}{x+5}}$.

Step 2. Horizontal compression by factor 3: replace x by 3x: $y = \sqrt{\frac{2(3x)+3}{3x+5}} = \sqrt{\frac{6x+3}{3x+5}}$.

Step 3. Vertical compression by factor 2: multiply by ½: $y = \frac{1}{2}\sqrt{\frac{6x+3}{3x+5}}$.

Step 4. Shift 1 unit downward: subtract 1: $y = \frac{1}{2}\sqrt{\frac{6x+3}{3x+5}} - 1$.

**Answer:** $y = \frac{1}{2}\sqrt{\frac{6x+3}{3x+5}} - 1$.

**Q8**

Question. Given the function f(x) = 3^x and its transformed function $g(x) = 3^{x-2} + 1$, draw the graphs of both functions on the same coordinate plane and compare their shapes.

Solution:

Step 1. Identify the transformations: $g(x) = 3^{x-2} + 1 = f(x-2) + 1$, so g is the graph of f shifted 2 units right and 1 unit up.

Step 2. Key points of f(x) = 3^x: (-1, ¹/₃), (0, 1), (1, 3), (2, 9); horizontal asymptote y = 0.

Step 3. Key points of g (shift each point of f right 2, up 1): (-1+2, ¹/₃+1) = (1, ⁴/₃); (0+2, 1+1) = (2, 2); (1+2, 3+1) = (3, 4); (2+2, 9+1) = (4, 10). Check directly: g(2) = $3^0$ + 1 = 2 ✓; g(0) = $3^{-2}$ + 1 = ¹/₉ + 1 = ¹⁰/₉ ✓.

Step 4. Asymptote of g: the line y = 0 shifts up 1 → y = 1.

Step 5. Comparison: both curves have exactly the same shape (shifting never changes shape) — g is f moved right 2 and up 1; g grows at the same rate, just from a higher, right-shifted position.

**Answer: $g(x) = 3^{x-2} + 1$ is f(x) = 3^x shifted 2 units right and 1 unit up; identical shape, asymptote moves from y = 0 to y = 1.**

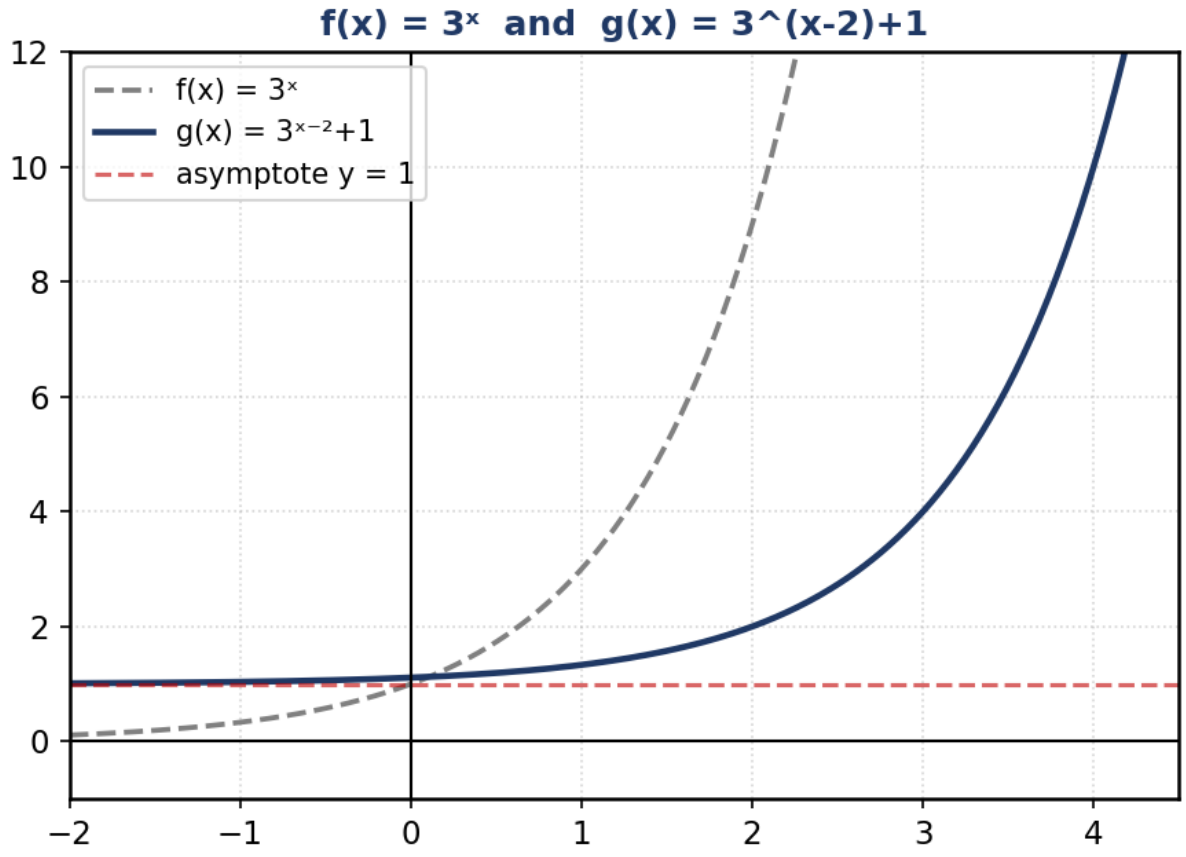



Fig: g(x) = 3^(x-2)+1 is f(x) = 3ˣ shifted 2 right and 1 up; same shape.

**Q9**

Question. Given the function $f(x) = \ln x$. Find the equation of the transformed function after applying in order: the vertical stretch by factor 4, shift 6 units to right, horizontal compress by factor 3, and shift 1 unit upward. Draw the graphs of both the original and transformed functions on the same coordinate plane.

Solution:

Step 1. Vertical stretch by factor 4: $y = 4 \cdot \ln x$.

Step 2. Shift 6 units right: replace x by x - 6: $y = 4 \cdot \ln(x - 6)$.

Step 3. Horizontal compression by factor 3: replace x by 3x: $y = 4 \cdot \ln(3x - 6)$.

Step 4. Shift 1 unit upward: y = $4 \cdot \ln(3x - 6) + 1$.

Step 5. Features for the drawing: original $f(x) = \ln x$ — vertical asymptote x = 0, passes through (1, 0), strictly increasing. Transformed $g(x) = 4 \cdot \ln(3x - 6) + 1$ — domain: 3x - 6 > 0 → x > 2; vertical asymptote x = 2; passes through (1, 0)-shifted point: $g(\frac{7}{3}) = 4 \cdot \ln(1) + 1 = 1$, i.e. (7/3, 1); the $\times 4$ stretch makes it rise 4 times faster vertically; still strictly increasing.

Step 6. Comparison: same basic log shape, but steeper (vertical stretch), moved right so its asymptote is x = 2, and lifted 1 unit up.

**Answer: $g(x) = 4 \cdot \ln(3x - 6) + 1$; same log shape, vertically stretched $\times 4$, asymptote shifted from x = 0 to x = 2, shifted up 1.**

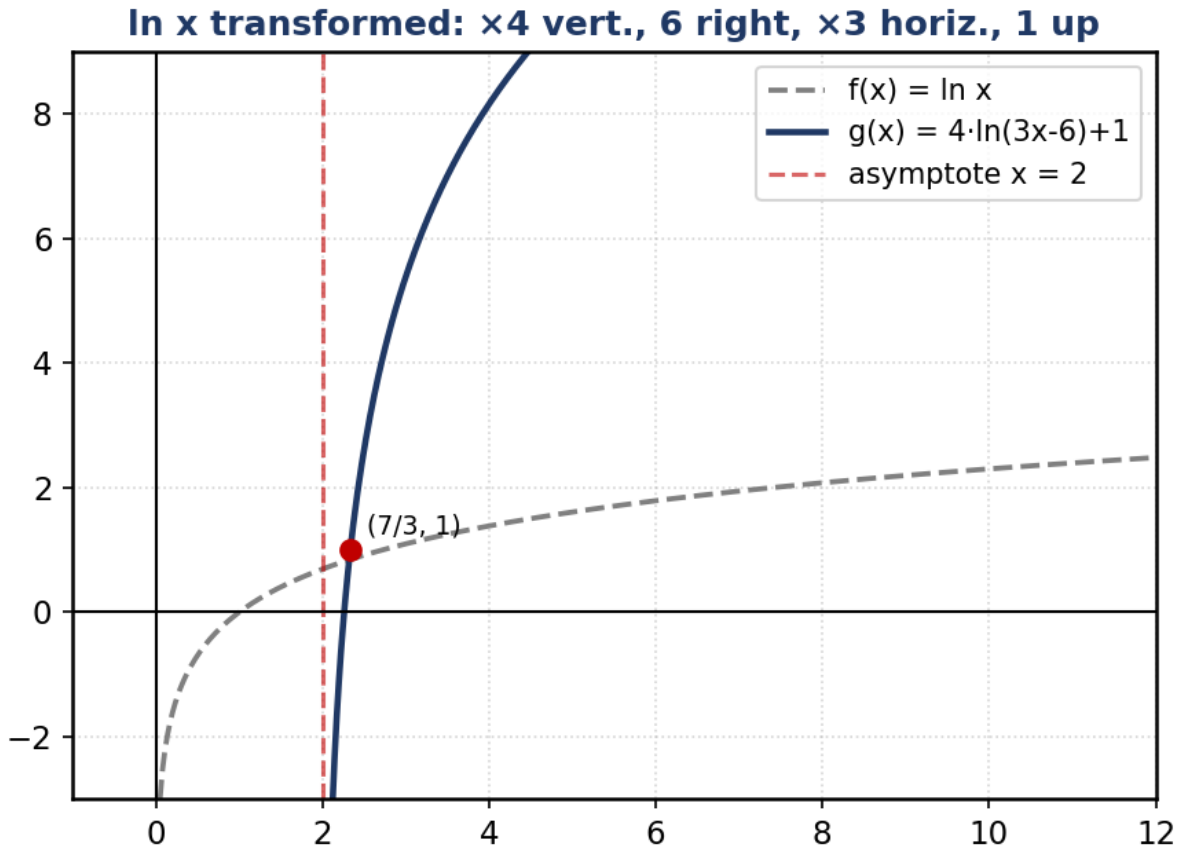



Fig: g(x) = 4·ln(3x-6)+1: ln x with ×4 vertical stretch, 6 right, ×3 horizontal compression, 1 up; asymptote moves from x = 0 to x = 2.

**Q10**

Question. Draw the graph of f(x) = |x| after applying in order: shift of 7 units to right, horizontal compress by factor 2, vertical shift 6 units downward, and vertical compress by factor 2.

Solution:

Step 1. Shift 7 units right: $y = |x - 7|$.

Step 2. Horizontal compression by factor 2: replace x by 2x: $y = |2x - 7|$.

Step 3. Vertical shift 6 units downward: $y = |2x - 7| - 6$.

Step 4. Vertical compression by factor 2: $y = \frac{1}{2}|2x - 7| - 6$.

Step 5. Features for the drawing: the V's corner (vertex) is where $2x - 7 = 0 \to x = \frac{7}{2} = 3.5$, and $y = -6 \to$ vertex (3.5, -6). The V is narrower than |x| (horizontal compression) and flatter (vertical compression by 2 halves the slope). Sample points: (0, $\frac{1}{2}(7) - 6 = -2.5$), (7, $\frac{1}{2}(7) - 6 = -2.5$), (3.5, -6).

**Answer: $y = \frac{1}{2}|2x - 7| - 6$; V-shape with vertex at (3.5, -6), narrower and vertically flattened compared to |x|.**

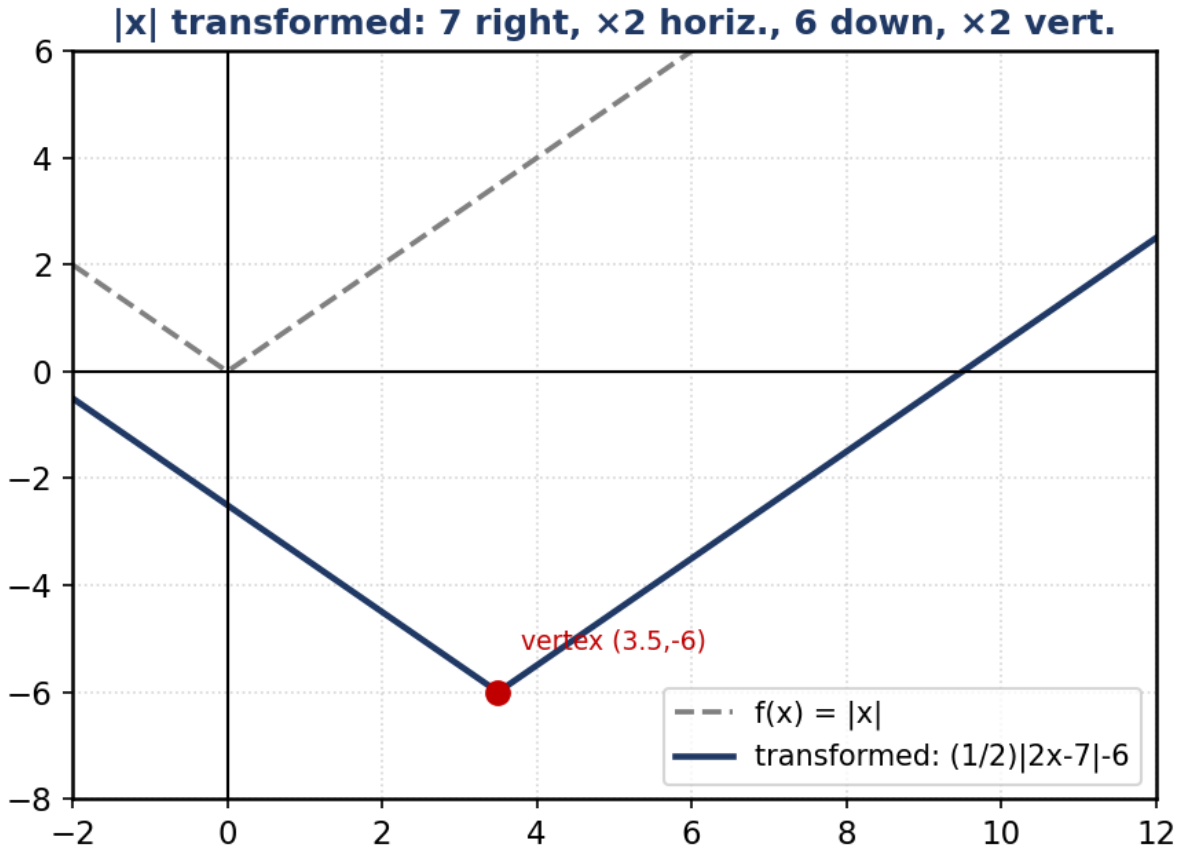



Fig: |x| transformed: 7 right, horizontal compression ×2, 6 down, vertical compression ×2.

## Notes on ambiguous wordings (for the teacher)

Step 1. Exercise 1.1, Q4 — "cuts the y-axis at y = -2" was read as "the y-intercept is (0, -2)". If instead it meant something else, the setup $y = a(x+1)^2$ with the given point would change.

Step 2. Exercise 1.1, Q2 — the book gives the graphs as figures; the descriptions used here are: (i) downward parabola, x-intercepts (-2, 0) and (1, 0), y-intercept (0, 2); (ii) upward parabola touching the x-axis at (2, 0), passing through (1, 1).

Step 3. Exercise 1.3, Q1(iv) and Q2(ii)–(iii), Q3 — "verify/draw graphically" parts are answered by describing the plotted points and the reflection construction, since the final PDF will carry the actual figures.

Step 4. Section 1.10 — the source PDF ends in the middle of Example 19, so 1.10 is presented as method notes only (no book examples copied). In the book's printed domain for Example 19 ("$\frac{1}{2} \leq x \leq \frac{5}{3}$"), the inequalities should strictly be x > ½ and x < ⅝, because a logarithm's argument must be strictly positive.

Step 5. Exercise 1.2, Q6(iii)–(iv) — these two parts print at the top of unit page 15 in the source; both are included above.